Muzaddidul Islam Chowdhury



I want to research how AI and analytics can make manufacturing supply chains more stable, especially in volatile, import-dependent economies.

Open to funded PhD positions for 2027 and 2028 intakes. plabon.mic@gmail.com | linkedin.com/in/muzaddidul | scholar.google.com/citations?user=mdswSJ8AAAAJ

2018–19 **Lab** — $BiFeO_3$ nanoparticles for solar cells

2018–21 **Devices** — Carbon-nanotube FET modelling, 2 IEEE papers

2021–22 **Storage** — MXene battery-electrode review

2021–23 **Factory** — Process Analyst, production engineering at BAT

2023–25 **Supply network** — Scheduler for a USD 67M+ material portfolio at BAT

2025–now **AI for manufacturing** — Project Manager at Aivelon

Four years at British American Tobacco Bangladesh took me from spare parts for the production line, with 8,000+ part types, to scheduling a USD 67M+ packaging-material portfolio across 10+ foreign suppliers. I built the tools I needed along the way: an Excel VBA MRP system, Power BI safety-stock models, and a Power Apps currency-risk app used through the 2023 forex crisis. Today I lead generative-AI and SaaS delivery for manufacturing clients at Aivelon. My electrical engineering degree and five publications give me the quantitative and research grounding.

> USD 67M+ material portfolio planned at 85% forecast accuracy.

> Measured results: 88 to 71 days lead time, 16% lower transport cost, 20% fewer write-offs, about £200K Lean savings.

> Now leading AI software delivery for manufacturers; 5 peer-reviewed engineering publications.

What I want to research

1. **Taming the bullwhip in shared raw materials**

   Managing plug-wrap paper shared across many SKUs, I watched small demand shifts cascade into large inventory swings and a near stock-out. I want to study forecasting and order-smoothing policies, including explainable ML, that planners will trust and use.

2. **Risk-aware procurement under currency and import shocks**

   Building LC Analyzer during the 2023 forex crisis showed me how currency, banking and lead-time risk interact for import-dependent plants. I want to model these jointly and turn them into procurement decisions.

3. **Generative AI that operations teams adopt**

   At Aivelon I see what manufacturers will and won't adopt. I want to research how AI copilots for planners and engineers in mid-size plants should be designed, measured and trusted.

Industry experience

Project Manager — Nov 2025 – present

Aivelon, Dhaka

Software company building SaaS and generative-AI-as-a-service products, currently focused on manufacturing clients.

- Lead the delivery team and client relationships, turning manufacturing clients' operational problems into software requirements and delivery plans.
- Supervise an engineering intern and work through the backend architecture (Java, Spring Boot) with him.

Wrapping Material Scheduler, Supply Network Operations Mar 2023 – Oct 2025
British American Tobacco Bangladesh, Dhaka

Planned packaging materials for a plant producing about 60 billion sticks a year.

- Built and ran inventory models for a USD 67M+ material portfolio, reaching 85% forecast accuracy from demand and consumption data.
- Coordinated 10+ foreign suppliers across Asia and Europe, from import documents and letters of credit to port clearance.
- Built *LC Analyzer* in Power Apps to track foreign-currency exposure during the 2023 forex crisis; every letter of credit for material imports was executed.
- Cut transport cost 16% with EOQ-based ordering and write-offs 20% with FIFO and shelf-life analysis.
- Applied Lean tools (value stream mapping, loss deployment) across the supply network, worth about £200K in savings.

Process Analyst, Production Engineering Jun 2021 – Feb 2023
British American Tobacco Bangladesh, Dhaka

Spare parts and maintenance readiness for the production line.

- Reduced overseas OEM spare-parts lead time from 88 to 71 days, improving maintenance readiness by 18%.
- Automated material requirement planning in Excel VBA, reaching 100% procurement accuracy.
- Built a Power BI model over 8,000+ spare-part types to set safety stock from consumption patterns at a 95% service level.
- Designed a technical training curriculum that brought a new team to full competency in six months.

Maintenance Engineer Oct 2019 – Jul 2021
Apsara Trading Ltd., Dhaka

Tools I built in industry

- **LC Analyzer.** Power Apps tool that scored letter-of-credit and currency risk for every import order during Bangladesh's 2023 forex crisis.
- **Project Amazon.** Power Apps digital indent system that replaced paper approvals and cut approval cycles by 85%.
- **Project Hulk.** Excel decision-support tool using lead-time analytics to set min-max levels and raise stock-out alerts automatically.
- **Automated MRP.** Excel VBA material requirement planning for spare parts, removing manual ordering errors.

Publications

MXene based heterostructures for electrode materials of batteries: a review
M. A. H. Shanto, **M. I. Chowdhury**, A. B. Antu, N. R. Niloy, N. Alam, M. A. Ullah, M. S. Anowar
IOP Conference Series: Materials Science and Engineering, vol. 1225, 012018 (IC-MAME 2021)
Open access
Journal-indexed proceedings, 2022 DOI

A study of OFF-state current and impact of dielectric constant on I_{ON}/I_{OFF} ratio of carbon nanotube field effect transistor
M. A. H. Shanto, **M. I. Chowdhury**, N. R. Niloy, N. Alam, M. J. Nayem, S.-A. Imam
2021 6th International Conference for Convergence in Technology (I2CT), Pune, India. IEEE
IEEE conference, 2021 DOI PDF

Multiferroic bismuth ferrite nanocomposites as a potential photovoltaic material
N. R. Niloy, **M. I. Chowdhury**, M. A. H. Shanto, J. Islam, M. M. Rhaman
IOP Conference Series: Materials Science and Engineering, vol. 1091, 012049 (ICIRMCT 2021)
Journal-indexed proceedings, 2021 DOI PDF

Structural and optical characterization of multiferroic $BiFeO_3$ nanoparticles synthesized at different annealing temperatures
N. R. Niloy, **M. I. Chowdhury**, S. Anowar, J. Islam, M. M. Rhaman
Journal of Nano- and Electronic Physics, vol. 12, no. 5, 05015 (2020)
Journal article, 2020 DOI PDF

Effect of channel length and dielectric constant on carbon nanotube FET to evaluate the device performance
M. A. H. Shanto, N. R. Niloy, R. Islam, **M. I. Chowdhury**, M. M. Abrar, S.-A. Imam
2020 4th International Conference on Electronics, Materials Engineering & Nano-Technology (IEMENTech), Kolkata, India, pp. 1–4. IEEE
IEEE conference, 2020 DOI PDF

Education

B.Sc. in Electrical and Electronic Engineering 2015 – 2021
Ahsanullah University of Science and Technology (AUST), Dhaka
Major in electronics. Thesis on bismuth ferrite nanoparticles for photovoltaics.

Skills

Planning and operations
SAP ERP, Kinaxis RapidResponse, Project44, FourKites; MRP, EOQ, min-max and safety stock; Lean (VSM, loss deployment)

Analytics and automation
Python, Power BI, Power Apps, Excel VBA and macros, MATLAB

Software delivery
Project management for SaaS and generative-AI products; working knowledge of Java and Spring Boot